BIOLOGY | NEET UG STANDARD | HUMAN REPRODUCTION & WELFARE

Reproductive Health

A comprehensive, self-contained study guide and assessment module covering national reproductive health strategies, population stabilisation, contraception, medical termination of pregnancy (MTP), sexually transmitted infections (STIs), and assisted reproductive technologies (ART).

1. Learning Objectives & Chapter Overview

In general biological terms, **reproductive health** simply refers to healthy reproductive organs with normal functions. However, in clinical and public-health contexts, the term carries a broader perspective that includes the emotional and social aspects of reproduction.

> **World Health Organisation (WHO) Definition of Reproductive Health**
>
> According to the World Health Organisation (WHO), **reproductive health** means a total well-being in all aspects of reproduction, i.e., **physical, emotional, behavioural and social**. Therefore, a society with people having physically and functionally normal reproductive organs and normal emotional and behavioural interactions among them in all sex-related aspects is called a reproductively healthy society.

Key Learning Objectives for NEET UG:

- Understand the historical timeline and scope of India's Family Planning and Reproductive and Child Health Care (RCH) programmes.
- Evaluate the purpose and legal status of **amniocentesis**, distinguishing its legitimate diagnostic use from its illegal misuse for sex determination.
- Analyse demographic trends in global and Indian population growth and identify the drivers of population explosion.
- Compare all classes of contraceptives (Natural, Barrier, Intra Uterine Devices, Oral Pills, Injectables/Implants, Emergency, and Surgical methods) by their exact composition and mechanism of action.
- Understand the clinical timeframes, legal provisions, and grounds for **Medical Termination of Pregnancy (MTP)**, including the 2017 Amendment Act.
- Recognise common **Sexually Transmitted Infections (STIs)**, distinguish curable from incurable infections, and identify complications of untreated STIs.
- Select appropriate **Assisted Reproductive Technologies (ART)**—such as IVF-ET, ZIFT, IUT, GIFT, ICSI, and AI/IUI—based on specific male or female infertility factors.

2. Core Concepts & Detailed Explanations

2.1 Reproductive Health — Problems and Strategies

India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal:

- **Family Planning Programmes (1951):** Initiated in 1951 and periodically assessed over the past decades.
- **Reproductive and Child Health Care (RCH) Programmes:** Improved programmes currently in operation covering wider reproduction-related areas. Their two major tasks are:
  1. Creating awareness among people about various reproduction-related aspects.
  2. Providing facilities and support for building up a reproductively healthy society.

Strategies for Creating Awareness

- **Media & Community Support:** Governmental and non-governmental agencies use audio-visual and print media to create awareness. Parents, other close relatives, teachers, and friends also play a major role in

disseminating accurate information.

- **Sex Education in Schools:** Encouraged to provide right information to the young so as to discourage children from believing in myths and having misconceptions about sex-related aspects. Proper knowledge of reproductive organs, adolescence and related changes, safe and hygienic sexual practices, sexually transmitted diseases (STD), and AIDS helps adolescents lead a reproductively healthy life.
- **Educating Fertile Couples & Marriageable Age Groups:** Educating people about available birth control options, care of pregnant mothers, post-natal care of the mother and child, importance of breast-feeding, and equal opportunities for the male and the female child addresses the importance of bringing up socially conscious healthy families of desired size.
- **Social Awareness:** Creating awareness about problems due to uncontrolled population growth and social evils like sex-abuse and sex-related crimes enables people to think and take necessary steps to prevent them.

Infrastructure, Research, and Legal Safeguards

Successful implementation of action plans requires strong infrastructural facilities, professional expertise, and material support to provide medical assistance in pregnancy, delivery, STDs, abortions, contraception, menstrual problems, and infertility.

- **Amniocentesis and the Statutory Ban:** In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse the foetal cells and dissolved substances.
  - *Legitimate Medical Use:* Used to test for the presence of certain genetic disorders such as **Down syndrome, haemophilia, sickle-cell anemia**, etc., and to determine the survivability of the foetus.
  - *Reason for Statutory Ban:* A statutory ban on amniocentesis for **sex-determination** was enacted in India to legally check the increasing menace of female foeticides.
- **Massive Child Immunisation:** Another major national programme operating alongside RCH.
- **Indigenous Research Achievement — 'Saheli':** Research on reproduction-related areas is supported by governmental and non-governmental agencies. **'Saheli'**, a new oral contraceptive for females, was developed by scientists at the **Central Drug Research Institute (CDRI) in Lucknow, India**.
- **Indicators of Improved Reproductive Health:** Better awareness about sex-related matters, an increased number of medically assisted deliveries, better post-natal care leading to decreased maternal and infant mortality rates, an increased number of couples with small families, better detection and cure of STDs, and overall increased medical facilities for all sex-related problems all indicate improved reproductive health of the society.

2.2 Population Stabilisation and Birth Control

In the last century, all-round development in various fields significantly improved the quality of life of the people. However, increased health facilities along with better living conditions had an explosive impact on the growth of population.

Year / Period	World Population	Indian Population
1900	Around 2 billion (2000 million)	–
1947 (Time of Independence)	–	Approximately 350 million
2000	About 6 billion	Close to the 1 billion mark
May 2011	7.2 billion	Crossed 1.2 billion

Probable Reasons for Population Explosion

1. A rapid decline in **death rate**.
2. A rapid decline in **maternal mortality rate (MMR)**.
3. A rapid decline in **infant mortality rate (IMR)**.
4. An increase in the number of people in the **reproducible age**.

BOSONANCE

Through the RCH programme, although the population growth rate was brought down, the reduction was only marginal. According to the **2011 census report**, the population growth rate was less than 2%, i.e., $20/1000/\text{year}$, a rate at which our population could increase rapidly, threatening an absolute scarcity of basic requirements (food, shelter, and clothing).

Measures to Check Population Growth Rate

- **Motivating Smaller Families:** Promoting contraceptive use through media advertisements, posters, and bills showing a happy couple with two children with the slogan *Hum Do Hamare Do* (we two, our two). Many young, urban, working couples have even adopted a 'one child norm'.
- **Statutory Raising of Marriageable Age:** Raising the legal marriageable age of **females to 18 years** and that of **males to 21 years**.
- **Incentives:** Providing incentives to couples with small families.

2.3 Contraceptive Methods

> ### Characteristics of an Ideal Contraceptive
>
> An ideal contraceptive should be **user-friendly, easily available, effective and reversible with no or least side-effects**. It also should in no way interfere with the sexual drive, desire and/or the sexual act of the user.

Presently available contraceptive methods are broadly grouped into: **Natural/Traditional, Barrier, IUDs, Oral contraceptives, Injectables, Implants, and Surgical methods**.

A. Natural / Traditional Methods

These work on the principle of avoiding chances of ovum and sperms meeting. Because no medicines or devices are used, side effects are almost nil, though chances of failure of this method are high.

1. **Periodic Abstinence:** Couples avoid or abstain from coitus from **day 10 to 17 of the menstrual cycle** when ovulation could be expected. Because chances of fertilisation are very high during this period, it is called the **fertile period**.
2. **Withdrawal or *Coitus Interruptus*:** The male partner withdraws his penis from the vagina just before ejaculation so as to avoid insemination.
3. **Lactational Amenorrhea (Absence of Menstruation):** Based on the physiological fact that ovulation and therefore the menstrual cycle do not occur during the period of intense lactation following parturition. As long as the mother breast-feeds the child fully, chances of conception are almost nil. However, this method is effective only up to a **maximum period of six months** following parturition.

B. Barrier Methods

In barrier methods, ovum and sperms are prevented from physically meeting with the help of barriers. Available for both males and females:

- **Condoms:** Barriers made of thin rubber/latex sheath used to cover the penis in the male or vagina and cervix in the female just before coitus so that ejaculated semen does not enter the female reproductive tract.
  - **'Nirodh'** is a popular brand of condom for the male.
  - *Additional Benefit:* Protects the user from contracting STIs and AIDS.
  - Both male and female condoms are **disposable**, can be self-inserted, and thereby give privacy to the user.
- **Diaphragms, Cervical Caps, and Vaults:** Barriers made of rubber inserted into the female reproductive tract to cover the cervix during coitus. They prevent conception by blocking the entry of sperms through the cervix. Unlike condoms, they are **reusable**.
- **Spermicidal Creams, Jellies, and Foams:** Usually used along with these barriers to increase their contraceptive efficiency.

BOSONANCE

C. Intra Uterine Devices (IUDs)

Inserted by doctors or expert nurses in the uterus through the vagina. IUDs are ideal contraceptives for females who want to delay pregnancy and/or space children, and represent one of the most widely accepted methods of contraception in India.

- **Types of IUDs:**
  1. **Non-medicated IUDs:** e.g., *Lippes loop*.
  2. **Copper-releasing IUDs:** e.g., *CuT, Cu7, Multiload 375*.
  3. **Hormone-releasing IUDs:** e.g., *Progestasert, LNG-20*.

- **Mechanisms of Action:**
  - All IUDs **increase phagocytosis of sperms** within the uterus.
  - Copper ions (Cu ions) released **suppress sperm motility and the fertilising capacity of sperms**.
  - Hormone-releasing IUDs, in addition, **make the uterus unsuitable for implantation and the cervix hostile to the sperms**.

D. Oral Contraceptives (Pills)

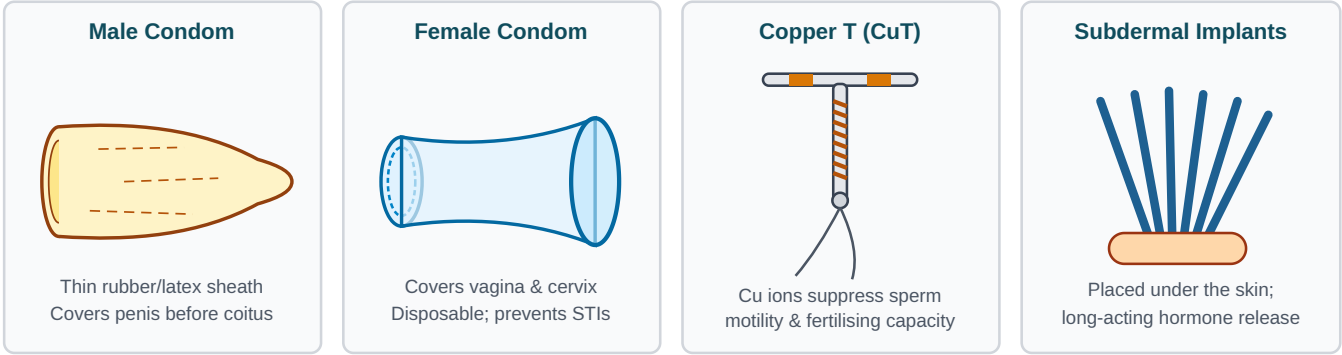
- **Hormonal Pills:** Oral administration of small doses of either **progestogens or progestogen–estrogen combinations** used by females in the form of tablets.
  - *Dosage Regimen:* Taken daily for a period of **21 days** starting preferably within the first five days of the menstrual cycle. After a gap of **7 days** (during which menstruation occurs), it has to be repeated in the same pattern till the female desires to prevent conception.
  - *Mechanism of Action:* They **inhibit ovulation and implantation** as well as **alter the quality of cervical mucus** to prevent/retard entry of sperms.

- **'Saheli':** Developed at CDRI, Lucknow. Contains a **non-steroidal preparation**. It is a **'once a week'** pill with very few side effects and high contraceptive value.

E. Injectables, Implants, and Emergency Contraceptives

- **Injectables and Implants:** Progestogens alone or in combination with estrogen can also be used by females as injections or implants under the skin. Their mode of action is similar to that of pills, and their effective periods are much longer.
- **Emergency Contraceptives:** Administration of **progestogens or progestogen–estrogen combinations or IUDs within 72 hours of coitus** have been found to be very effective as emergency contraceptives to avoid possible pregnancy due to rape or casual unprotected intercourse.

Tutorial Diagram 1: Representative Contraceptive Devices (Barriers, IUD, and Subdermal Implants)



Comparison of barrier sheaths (male and female condoms), a copper-releasing intra-uterine device (CuT), and hormone-releasing subdermal implants.

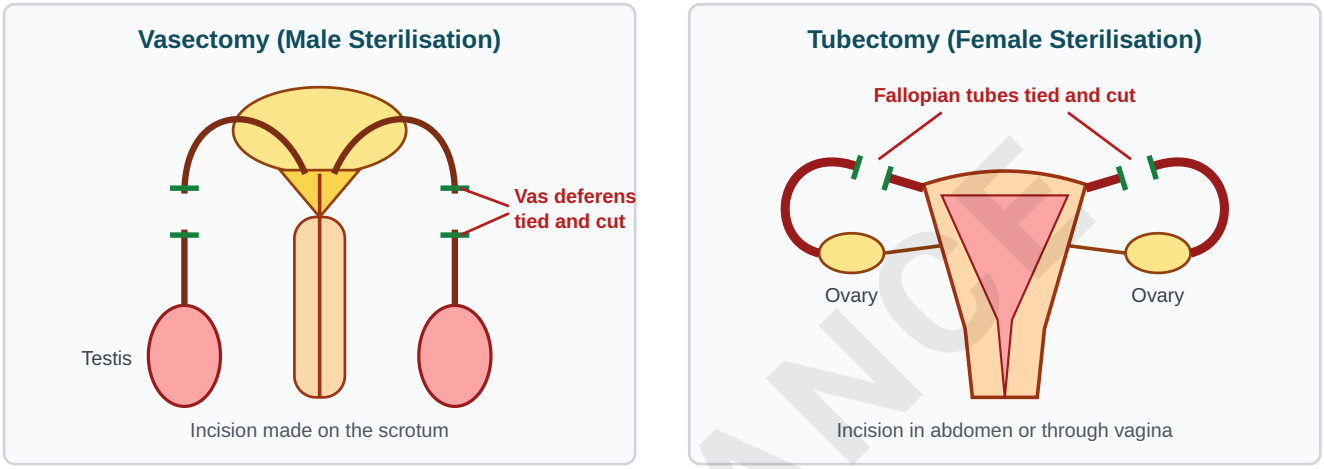
BOSONANCE

F. Surgical Methods (Sterilisation)

Advised for the male/female partner as a **terminal method** to prevent any more pregnancies. Surgical intervention **blocks gamete transport** and thereby prevents conception (note: gametogenesis and hormone production remain unaffected). These techniques are highly effective, but their **reversibility is very poor**.

- **Vasectomy (in Males):** A small part of the **vas deferens** is removed or tied up through a small incision on the scrotum.
- **Tubectomy (in Females):** A small part of the **fallopian tube** is removed or tied up through a small incision in the abdomen or through the vagina.

Tutorial Diagram 2: Surgical Methods of Contraception (Sterilisation)



Surgical sterilisation blocks gamete transport without stopping gamete formation: Vasectomy cuts and ties the vas deferens; Tubectomy cuts and ties the fallopian tubes.

Precautions and Possible Ill-Effects of Contraceptives

Selection of a suitable contraceptive method and its use should always be undertaken in consultation with qualified medical professionals. Contraceptives are not regular requirements for the maintenance of reproductive health; they are practiced against a natural reproductive event (conception/pregnancy). Possible ill-effects include **nausea, abdominal pain, breakthrough bleeding, irregular menstrual bleeding, or even breast cancer**, which, though not very significant, should not be totally ignored.

2.4 Medical Termination of Pregnancy (MTP)

Intentional or voluntary termination of pregnancy before full term is called **medical termination of pregnancy (MTP)** or **induced abortion**.

- **Global Statistics:** Nearly $45 \text{ to } 50 \text{ million}$ **MTPs** are performed in a year all over the world, which accounts for $1/5\text{th}$ **of the total number of conceived pregnancies** in a year.
- **Legalisation in India:** Debated globally due to emotional, ethical, religious, and social issues. The **Government of India legalised MTP in 1971** with strict conditions to avoid its misuse and to check indiscriminate and illegal female foeticides.
- **Why MTP is Performed:** To get rid of unwanted pregnancies either due to casual unprotected intercourse, failure of the contraceptive used during coitus, or rapes. Also essential in cases where continuation of the pregnancy could be harmful or even fatal either to the mother or to the foetus or both.
- **Safe Window:** MTPs are considered relatively safe during the **first trimester, i.e., up to 12 weeks of pregnancy**. Second trimester abortions are much riskier.
- **Disturbing Trends:**
  1. A majority of MTPs are performed illegally by unqualified quacks, which are unsafe and potentially fatal.

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2. Misuse of amniocentesis to determine the sex of the unborn child, followed by MTP if the foetus is found to be female (totally illegal).

The Medical Termination of Pregnancy (Amendment) Act, 2017

Enacted by the Government of India with the intention of reducing the incidence of illegal abortion and consequent maternal mortality and morbidity:

- **Within the first 12 weeks of pregnancy:** May be terminated on the opinion of **one registered medical practitioner**.
- **More than 12 weeks, but fewer than 24 weeks: Two registered medical practitioners** must be of the opinion, formed in good faith, that the required grounds exist.
- **Permitted Grounds for Termination:**
  1. The continuation of the pregnancy would involve a risk to the life of the pregnant woman or of grave injury to her physical or mental health; OR
  2. There is a substantial risk that if the child were born, it would suffer from such physical or mental abnormalities as to be seriously handicapped.

2.5 Sexually Transmitted Infections (STIs)

Infections or diseases transmitted through sexual intercourse are collectively called **sexually transmitted infections (STI)** or **venereal diseases (VD)** or **reproductive tract infections (RTI)**.

- **Common Examples:** Gonorrhoea, syphilis, genital herpes, chlamydiasis, genital warts, trichomoniasis, hepatitis-B, and HIV leading to AIDS (the most dangerous STI).
- **Additional Modes of Transmission (for Hepatitis-B and HIV):**
  1. Sharing of injection needles, surgical instruments, etc., with infected persons.
  2. Transfusion of blood.
  3. From an infected mother to the foetus.
- **Curability Rule (High-Yield for NEET):** Except for **hepatitis-B, genital herpes, and HIV infections**, other STIs are completely curable if detected early and treated properly.
- **Early Symptoms & Detection Challenges:** Early symptoms are minor and include itching, fluid discharge, slight pain, swellings, etc., in the genital region. Infected females may often be **asymptomatic** and hence remain undetected for long. Absence or less significant early symptoms and social stigma deter infected persons from timely detection and proper treatment.
- **Complications of Untreated STIs: Pelvic inflammatory diseases (PID), abortions, still births, ectopic pregnancies, infertility, or even cancer of the reproductive tract**.
- **High-Vulnerability Age Group:** Though all persons are vulnerable, incidences are reported to be very high among persons in the **age group of 15–24 years**.
- **Three Principles of Prevention:**
  1. Avoid sex with unknown partners/multiple partners.
  2. Always try to use condoms during coitus.
  3. In case of doubt, go to a qualified doctor for early detection and get complete treatment if diagnosed with infection.

2.6 Infertility and Assisted Reproductive Technologies (ART)

Infertility is defined as the inability to conceive or produce children even after **2 years of unprotected sexual cohabitation**.

- **Causes:** Physical, congenital, diseases, drugs, immunological, or even psychological.
- **Social Misconception vs. Clinical Reality:** In India, often the female is blamed for the couple being childless, but more often than not, the problem lies in the male partner.
- **Role of Infertility Clinics & ART:** Specialised health care units (infertility clinics) help in diagnosis and corrective treatment of some disorders. Where such corrections are not possible, couples are assisted

through special techniques called **Assisted Reproductive Technologies (ART)**.

Classification of Assisted Reproductive Technologies (ART)

1. ***In Vitro* Fertilisation (IVF) followed by Embryo Transfer (ET) — 'Test Tube Baby' Programme:**
   - Ova from the wife/donor (female) and sperms from the husband/donor (male) are collected and induced to form a zygote under simulated conditions in the laboratory (outside the body in almost similar conditions as that in the body).
   - **ZIFT (Zygote Intra Fallopian Transfer):** The zygote or early embryos with **up to 8 blastomeres** are transferred into the **fallopian tube**.
   - **IUT (Intra Uterine Transfer):** Embryos with **more than 8 blastomeres** are transferred into the **uterus** to complete further development.
   - *Note on In-Vivo Fertilisation:* Embryos formed by ***in-vivo* fertilisation** (fusion of gametes within the female) can also be used for such transfer to assist those females who cannot conceive.
2. **GIFT (Gamete Intra Fallopian Transfer):** Transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce one, but can provide a suitable environment for fertilisation and further development.
3. **ICSI (Intra Cytoplasmic Sperm Injection):** A specialised procedure to form an embryo in the laboratory in which a sperm is directly injected into the ovum.
4. **AI (Artificial Insemination) & IUI (Intra-Uterine Insemination):** Used when infertility is due either to the inability of the male partner to inseminate the female or due to very low sperm counts in the ejaculates. Semen collected either from the husband or a healthy donor is artificially introduced either into the vagina or into the uterus (IUI) of the female.

Limitations of ART & Legal Adoption

ART techniques require extremely high precision handling by specialised professionals and expensive instrumentation, making them available in very few centres and affordable to only a limited number of people. Emotional, religious, and social factors are also deterrents. Since the ultimate aim is to have children and India has many orphaned and destitute children, **legal adoption** permitted by Indian laws is one of the best methods for couples looking for parenthood.

BOSONANCE

3. Master Reference Tables & Key Quantitative Facts

3.1 Comprehensive Comparison of Contraceptive Methods

Category	Specific Examples	Mechanism of Action & Key Features
Natural / Traditional	1. Periodic abstinence (days 10–17) 2. Withdrawal (*coitus interruptus*) 3. Lactational amenorrhea (up to 6 months post-parturition)	Avoids chances of ovum and sperms meeting. No medicines/devices used; almost nil side effects; high failure rate.
Barrier Methods	• Condoms (Male: *Nirodh*; Female) • Diaphragms, cervical caps, vaults (Female) • Spermicidal creams, jellies, foams	Prevent physical meeting of ovum and sperms. Condoms are disposable and protect against STIs/AIDS. Diaphragms, cervical caps, and vaults cover the cervix and are reusable.
Intra Uterine Devices (IUDs)	• Non-medicated: *Lippes loop* • Cu-releasing: *CuT, Cu7, Multiload 375* • Hormone-releasing: *Progestasert, LNG-20*	Increase phagocytosis of sperms in uterus. Cu ions suppress sperm motility and fertilising capacity. Hormone IUDs make uterus unsuitable for implantation and cervix hostile to sperms.
Oral Contraceptives (Pills)	• Steroidal pills (progestogens or progestogen–estrogen) • Non-steroidal pill: *Saheli* (CDRI, Lucknow)	Inhibit ovulation and implantation; alter cervical mucus to prevent/retard sperm entry. Standard pills: 21 days daily + 7-day gap. *Saheli*: once-a-week non-steroidal pill.
Injectables & Implants	Progestogens alone or with estrogen	Placed under skin or injected; mode of action similar to pills but with much longer effective periods.
Emergency Contraceptives	Progestogens, progestogen–estrogen combinations, or IUDs	Administered within **72 hours of coitus**; prevents pregnancy from rape or casual unprotected intercourse.
Surgical Methods (Sterilisation)	• Male: **Vasectomy** (vas deferens cut/tied) • Female: **Tubectomy** (fallopian tube cut/tied)	Terminal method; blocks gamete transport. Highly effective, but reversibility is very poor.

BOSONANCE

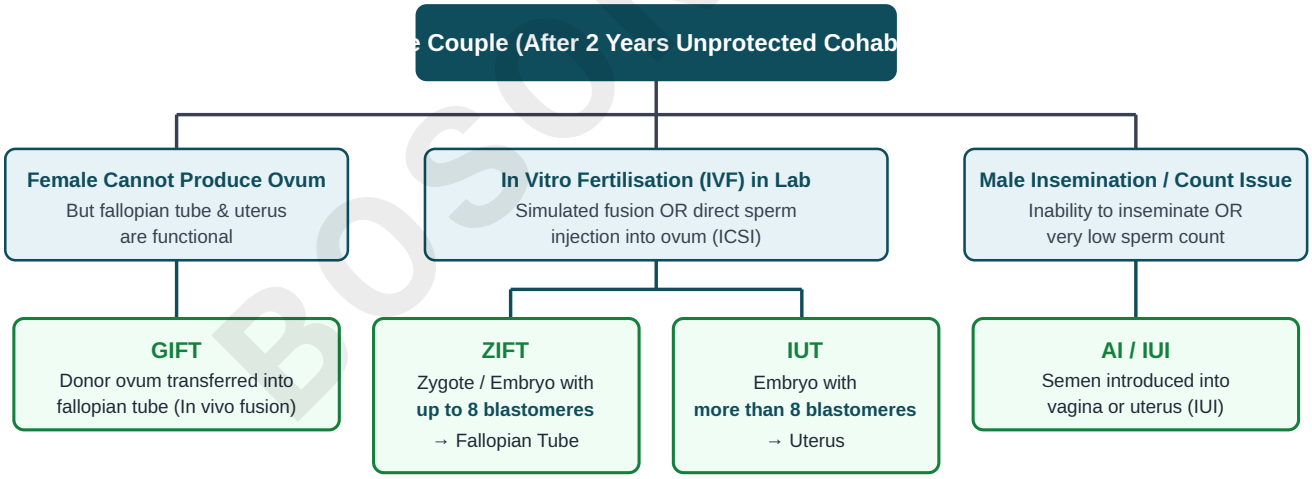
3.2 Assisted Reproductive Technologies (ART) Summary

Acronym & Full Name	Site of Fertilisation	What is Transferred & Where?	Primary Clinical Indication
ZIFT (Zygote Intra Fallopian Transfer)	*In vitro* (in lab)	Zygote or early embryo (**up to 8 blastomeres**) into the **fallopian tube**	Part of IVF–ET ('Test Tube Baby') programme
IUT (Intra Uterine Transfer)	*In vitro* (or *in vivo* donor embryo)	Embryo with **more than 8 blastomeres** into the **uterus**	Part of IVF–ET programme or for females who cannot conceive
GIFT (Gamete Intra Fallopian Transfer)	*In vivo* (in fallopian tube)	**Donor ovum** into the **fallopian tube** of recipient female	Female cannot produce ovum, but can provide suitable environment for fertilisation and development
ICSI (Intra Cytoplasmic Sperm Injection)	*In vitro* (in lab)	Single **sperm directly injected into the ovum** in the laboratory	Severe male factor infertility requiring direct micro-injection
AI / IUI (Artificial Insemination / Intra-Uterine Insemination)	*In vivo* (in female tract)	**Semen** (from husband or healthy donor) into **vagina or uterus (IUI)**	Inability of male partner to inseminate female OR very low sperm counts in ejaculate

4. Clinical Decision Pathways & Conceptual Connections

In NEET UG, questions frequently test your ability to select the right ART protocol based on which reproductive step is impaired, or to distinguish between where fertilisation occurs (*in vitro* vs. *in vivo*) and where the gamete or embryo is placed.

Tutorial Diagram 3: Clinical Decision Flowchart for Assisted Reproductive Technologies (ART)



Decision pathway for selecting GIFT, ZIFT, IUT, ICSI, or AI/IUI based on the specific cause of infertility and blastomere count.

5. Worked Conceptual Examples & Statement Analyses

The following worked examples address fundamental conceptual questions, clinical justifications, and statement-correction exercises essential for mastering this topic.

BOSONANCE

Worked Example 1: Significance of Reproductive Health & Priority Areas

Conceptual Analysis

Problem: (a) What is the significance of reproductive health in a society? (b) Which aspects of reproductive health need to be given special attention in the present scenario?

Solution:

(a) Significance in Society: Reproductive health ensures total well-being in physical, emotional, behavioural, and social aspects of reproduction. It prevents the spread of sexually transmitted infections (STIs) including AIDS, reduces maternal mortality rate (MMR) and infant mortality rate (IMR) through medically assisted deliveries and post-natal care, checks uncontrolled population growth through family planning, prevents social evils such as female foeticide and sex-abuse, and assists infertile couples to achieve parenthood.

(b) Priority Aspects in the Present Scenario:

- Counselling and creating awareness among adolescents about reproductive organs, puberty-related changes, and safe/hygienic sexual practices.
- Educating fertile couples and those of marriageable age about birth control options, pre-natal care of pregnant mothers, post-natal care, and the importance of breast-feeding.
- Preventing and ensuring early detection and cure of STIs (especially in the vulnerable 15–24 years age group).
- Enforcing strict legal checks against the misuse of amniocentesis for sex determination and illegal female foeticide.

Worked Example 2: Sex Education in Schools & 50-Year Progress in India

Public Health Evaluation

Problem: (a) Is sex education necessary in schools? Why? (b) Has reproductive health in India improved in the past 50 years? Mention specific areas of improvement.

Solution:

(a) Necessity of Sex Education in Schools: Yes, introduction of sex education in schools is necessary and should be encouraged to provide accurate, scientific information to the young. This discourages children and adolescents from believing in myths and having misconceptions about sex-related aspects. Proper information about reproductive organs, adolescence and related changes, safe and hygienic sexual practices, STDs, and AIDS helps adolescents lead a reproductively healthy life.

(b) Areas of Improvement in the Past 50 Years: Yes, reproductive health in India has improved significantly since family planning programmes were initiated in 1951. Key indicators of improvement include:

1. Better awareness about sex-related matters among the public.
2. An increased number of medically assisted deliveries and better post-natal care, leading to decreased maternal mortality rates (MMR) and infant mortality rates (IMR).
3. An increased number of couples adopting small family norms.
4. Better detection and cure of STIs and overall increased medical facilities for all sex-related problems (including infertility clinics).

BOSONANCE

Worked Example 3: Contraceptive Justification vs. Gonad Removal

Physiological Reasoning

Problem: (a) Is the use of contraceptives justified? Give reasons. (b) Why can the surgical removal of gonads (testes or ovaries) not be considered as a contraceptive option?

Solution:

(a) Justification for Contraceptive Use: Yes, the use of contraceptives is completely justified. Rapid declines in death rate, MMR, and IMR alongside an increased reproducible-age population caused an explosive growth of population (crossing $1.2\ \text{billion}$ in India in May 2011). Unchecked population growth leads to an absolute scarcity of basic requirements like food, shelter, and clothing. Furthermore, contraceptives help individuals avoid unwanted pregnancies, space or delay pregnancies for personal and maternal health reasons, and (in the case of condoms) protect against STIs and AIDS.

(b) Why Removal of Gonads is Not a Contraceptive Option: An ideal contraceptive must be user-friendly, easily available, effective, and **reversible** with minimal or no side effects, and must not interfere with the sexual drive or act. Surgical removal of gonads (castration/oophorectomy) is completely **irreversible** and permanently stops not only gamete production but also the secretion of essential sex hormones (androgens in males; estrogen and progesterone in females), leading to severe physiological and endocrine disturbances and loss of sexual drive. (By contrast, surgical sterilisation like vasectomy or tubectomy only blocks gamete transport while leaving the gonads intact.)

Worked Example 4: Verification of True/False Statements with Explanations

Statement Verification

Problem: State whether each of the following statements is **True** or **False**, and provide a scientific explanation for each:

a. Abortions could happen spontaneously too.

b. Infertility is defined as the inability to produce a viable offspring and is always due to abnormalities/defects in the female partner.

c. Complete lactation could help as a natural method of contraception.

d. Creating awareness about sex-related aspects is an effective method to improve reproductive health of the people.

Solution:

- **(a) True:** While Medical Termination of Pregnancy (MTP) is an intentional or voluntary induced abortion, abortions can also occur spontaneously (miscarriage) due to internal maternal, foetal, or pathological factors (such as untreated STIs).
- **(b) False:** Infertility is the inability to conceive or produce children even after 2 years of unprotected sexual cohabitation, and it is **not** always due to the female partner. Although females are often socially blamed in India, more often than not, the problem lies in the male partner (e.g., low sperm count or inability to inseminate).
- **(c) True:** This method is called **lactational amenorrhea**. During the period of intense lactation following parturition, ovulation and the menstrual cycle do not occur. However, it is effective only up to a maximum period of six months following parturition.
- **(d) True:** Creating awareness through audio-visual media, print media, teachers, and parents is the primary task under Reproductive and Child Health Care (RCH) programmes to eliminate myths, prevent STIs, and promote healthy family planning.

BOSONANCE

Worked Example 5: Correcting Common Misconceptions

Error Correction

Problem: Identify the scientific error in each of the following statements and rewrite them accurately:

a. Surgical methods of contraception prevent gamete formation.

b. All sexually transmitted diseases are completely curable.

c. Oral pills are very popular contraceptives among the rural women.

d. In E.T. techniques, embryos are always transferred into the uterus.

Solution:

- **(a) Corrected Statement:** Surgical methods of contraception (vasectomy and tubectomy) block **gamete transport** and thereby prevent conception; they do *not* prevent gamete formation.
- **(b) Corrected Statement:** Except for **hepatitis-B, genital herpes, and HIV infections**, other sexually transmitted diseases are completely curable if detected early and treated properly.
- **(c) Corrected Statement: Intra Uterine Devices (IUDs)** are one of the most widely accepted methods of contraception in India, whereas oral pills require strict daily compliance for 21 days (or weekly for *Saheli*) and are well accepted generally by educated/urban females.
- **(d) Corrected Statement:** In Embryo Transfer (ET) techniques, zygotes or early embryos with **up to 8 blastomeres** are transferred into the **fallopian tube (ZIFT)**, and embryos with **more than 8 blastomeres** are transferred into the **uterus (IUT)**.

6. NEET UG Practice Questions with Complete Worked Solutions

Note: All questions below are original practice questions designed strictly in accordance with the NEET UG examination pattern (Single-Correct MCQs, Multi-Statement Evaluation, Assertion–Reason, and Matrix-Match formats).

Practice Question 1

Single-Correct MCQ

Which of the following copper-releasing Intra Uterine Devices (IUDs) acts by releasing Cu ions to suppress sperm motility and the fertilising capacity of sperms?

(1) Lippes loop and Progestasert

(2) CuT, Cu7, and Multiload 375

(3) LNG-20 and Progestasert

(4) Lippes loop, Cu7, and LNG-20

Correct Answer: (2)

Worked Solution: Intra Uterine Devices are classified into three groups: non-medicated IUDs (*Lippes loop*), copper-releasing IUDs (*CuT, Cu7, Multiload 375*), and hormone-releasing IUDs (*Progestasert, LNG-20*). Only the copper-releasing IUDs release Cu ions that suppress sperm motility and fertilising capacity.

BOSONANCE

Practice Question 2

Single-Correct MCQ

Which of the following characteristics is **correct** regarding the oral contraceptive pill *'Saheli'*?

(1) It is a daily steroidal pill containing a high dose of progestogen–estrogen combination.

(2) It was developed at the Central Drug Research Institute (CDRI), Lucknow, contains a non-steroidal preparation, and is taken once a week.

(3) It is a hormone-releasing intra-uterine device inserted by a doctor or expert nurse.

(4) It is an emergency surgical implant effective only within 72 hours of unprotected coitus.

Correct Answer: (2)

Worked Solution: *'Saheli'* is a novel oral contraceptive for females developed by scientists at the Central Drug Research Institute (CDRI) in Lucknow, India. It contains a non-steroidal preparation and is a 'once a week' pill with very few side effects and high contraceptive value.

Practice Question 3

Multi-Statement Question

Read the following statements regarding the procedure of **amniocentesis**:

- **Statement I:** In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse foetal cells and dissolved substances.
- **Statement II:** It can be used to detect genetic disorders such as Down syndrome, haemophilia, and sickle-cell anemia.
- **Statement III:** It is legally used across India to determine the sex of the unborn child to promote family planning.

Which of the above statements are **correct**?

(1) Statements I and II only

(2) Statements II and III only

(3) Statements I and III only

(4) Statements I, II, and III

Correct Answer: (1)

Worked Solution: Statements I and II are accurate descriptions of amniocentesis and its legitimate medical application. Statement III is incorrect because there is a strict statutory ban on amniocentesis for sex-determination in India to legally check the menace of female foeticides.

BOSONANCE

Practice Question 4

Assertion–Reason

Assertion (A): Lactational amenorrhea is a natural contraceptive method that is reliable up to two full years after parturition.

Reason (R): Ovulation and therefore the menstrual cycle do not occur during the period of intense lactation following parturition.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

Correct Answer: (4)

Worked Solution: Assertion (A) is false because lactational amenorrhea has been reported to be effective only up to a **maximum period of six months** following parturition, not two years. Reason (R) is true: intense lactation prevents ovulation and the menstrual cycle.

Practice Question 5

Single-Correct MCQ

According to the Medical Termination of Pregnancy (Amendment) Act, 2017 enacted by the Government of India, if a pregnancy has lasted **more than 12 weeks, but fewer than 24 weeks**, how many registered medical practitioners must opine in good faith that the required legal grounds exist for termination?

(1) One registered medical practitioner

(2) Two registered medical practitioners

(3) Three registered medical practitioners

(4) No medical practitioner opinion is required before 24 weeks

Correct Answer: (2)

Worked Solution: Under the Medical Termination of Pregnancy (Amendment) Act, 2017, a pregnancy may be terminated within the first 12 weeks on the opinion of **one** registered medical practitioner. If the pregnancy has lasted more than 12 weeks, but fewer than 24 weeks, **two** registered medical practitioners must be of the opinion, formed in good faith, that the required grounds exist.

BOSONANCE

Practice Question 6

Single-Correct MCQ

Which set of Sexually Transmitted Infections (STIs) represents diseases that are **NOT** completely curable even if detected early?

(1) Gonorrhoea, Syphilis, and Chlamydiasis

(2) Hepatitis-B, Genital herpes, and HIV infection

(3) Trichomoniasis, Genital warts, and Syphilis

(4) Chlamydiasis, Hepatitis-B, and Gonorrhoea

Correct Answer: (2)

Worked Solution: Except for **hepatitis-B, genital herpes, and HIV infections**, all other common STIs (such as gonorrhoea, syphilis, chlamydiasis, genital warts, and trichomoniasis) are completely curable if detected early and treated properly.

BOSONANCE

Practice Question 7

Matching Type

Match the Assisted Reproductive Technology in **Column I** with its exact procedural description in **Column II**:

Column I (ART Technique)	Column II (Procedural Description)
A. ZIFT	(i) Transfer of an embryo with more than 8 blastomeres into the uterus
B. IUT	(ii) Direct injection of a sperm into an ovum in the laboratory
C. GIFT	(iii) Transfer of a zygote or early embryo (up to 8 blastomeres) into the fallopian tube
D. ICSI	(iv) Transfer of a donor ovum into the fallopian tube of a female who cannot produce one

(1) A–(iii), B–(i), C–(iv), D–(ii)

(2) A–(i), B–(iii), C–(iv), D–(ii)

(3) A–(iii), B–(iv), C–(i), D–(ii)

(4) A–(ii), B–(i), C–(iv), D–(iii)

Correct Answer: (1)

Worked Solution:

- **ZIFT (Zygote Intra Fallopian Transfer):** Zygote or early embryo with up to 8 blastomeres is transferred into the fallopian tube → (iii).
- **IUT (Intra Uterine Transfer):** Embryo with more than 8 blastomeres is transferred into the uterus → (i).
- **GIFT (Gamete Intra Fallopian Transfer):** Ovum collected from a donor is transferred into the fallopian tube of another female who cannot produce one → (iv).
- **ICSI (Intra Cytoplasmic Sperm Injection):** A sperm is directly injected into the ovum in the laboratory → (ii).

BOSONANCE

Practice Question 8

Single-Correct MCQ

Which of the following is the specific additional mechanism of action exhibited by **hormone-releasing IUDs** (such as *Progestasert* and *LNG-20*) compared to non-medicated IUDs?

(1) They physically cover the penis or vagina during coitus.

(2) They make the uterus unsuitable for implantation and the cervix hostile to the sperms.

(3) They permanently sever and tie the fallopian tubes.

(4) They stimulate intense lactation to prevent ovulation.

Correct Answer: (2)

Worked Solution: While all IUDs increase phagocytosis of sperms within the uterus and copper-releasing IUDs release Cu ions to suppress sperm motility, hormone-releasing IUDs (*Progestasert, LNG-20*) additionally make the uterus unsuitable for implantation and the cervix hostile to sperms.

Practice Question 9

Single-Correct MCQ

In which of the following clinical situations is **Artificial Insemination (AI) / Intra-Uterine Insemination (IUI)** most specifically indicated?

(1) When the female partner has bilaterally blocked fallopian tubes.

(2) When the female partner cannot produce ova due to ovarian failure.

(3) When infertility is due either to the inability of the male partner to inseminate the female or due to very low sperm counts in the ejaculates.

(4) When the female partner lacks a functional uterus for implantation.

Correct Answer: (3)

Worked Solution: Infertility cases caused either by the inability of the male partner to inseminate the female or by very low sperm counts in the ejaculates can be corrected by Artificial Insemination (AI), in which semen collected from the husband or a healthy donor is artificially introduced into the vagina or into the uterus (IUI) of the female.

BOSONANCE

Practice Question 10 Single-Correct MCQ

Examine the following demographic milestones regarding human population growth and select the **correct** combination:

(1) World population in 1900: $2\ \text{billion}$; India's population at independence: $\approx 350\ \text{million}$; India's growth rate (2011 census): less than 2% $(20/1000/\text{year})$.

(2) World population in 1900: $6\ \text{billion}$; India's population at independence: $1\ \text{billion}$; India's growth rate (2011 census): 5%.

(3) World population in 2000: $2\ \text{billion}$; India's population in May 2011: $350\ \text{million}$; India's growth rate (2011 census): 20%.

(4) World population in 2011: $6\ \text{billion}$; India's population in 2000: $350\ \text{million}$; Family planning initiated in: 1971.

Correct Answer: (1)

Worked Solution: World population was around $2\ \text{billion}$ $(2000\ \text{million})$ in 1900, rose to $6\ \text{billion}$ by 2000, and reached $7.2\ \text{billion}$ in 2011. India's population was approximately $350\ \text{million}$ at the time of independence, reached close to $1\ \text{billion}$ by 2000, and crossed $1.2\ \text{billion}$ in May 2011. According to the 2011 census report, India's population growth rate was less than 2%, i.e., $20/1000/\text{year}$.

BOSONANCE

Practice Question 11

Multi-Statement Question

Consider the following statements regarding barrier methods of contraception:

A. Condoms are barriers made of thin rubber/latex sheath used to cover the penis in the male or vagina and cervix in the female just before coitus.

B. Both male and female condoms are reusable for multiple cycles.

C. Diaphragms, cervical caps, and vaults are rubber barriers inserted into the female reproductive tract to cover the cervix during coitus, and they are reusable.

D. Use of condoms provides the additional benefit of protecting the user from contracting STIs and AIDS.

Which of the above statements are **correct**?

(1) A, B, and C only

(2) A, C, and D only

(3) B and D only

(4) A, B, C, and D

Correct Answer: (2)

Worked Solution: Statement B is incorrect because both male and female condoms are **disposable** (single-use), whereas diaphragms, cervical caps, and vaults are **reusable**. Statements A, C, and D are completely accurate.

BOSONANCE

Practice Question 12

Assertion–Reason

Assertion (A): Infected females suffering from Sexually Transmitted Infections (STIs) often develop severe late complications such as Pelvic Inflammatory Diseases (PID), ectopic pregnancies, still births, or infertility.

Reason (R): Infected females may often be asymptomatic in the early stages of STI infection and, combined with social stigma, may remain undetected and untreated for long.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

Correct Answer: (1)

Worked Solution: In STIs, early symptoms are minor and infected females may often be asymptomatic. Absence or less significant symptoms in the early stages and the social stigma attached to STIs deter infected persons from going for timely detection and proper treatment. This delay directly leads to late complications including PID, abortions, still births, ectopic pregnancies, infertility, or cancer of the reproductive tract.

Practice Question 13

Single-Correct MCQ

Which of the following regimens can be administered within **72 hours of coitus** as an effective **emergency contraceptive** to avoid possible pregnancy due to rape or casual unprotected intercourse?

(1) Periodic abstinence from day 10 to 17 of the menstrual cycle

(2) Administration of progestogens or progestogen–estrogen combinations or insertion of IUDs

(3) Use of spermicidal creams and jellies alone after 72 hours

(4) Performing amniocentesis within 72 hours

Correct Answer: (2)

Worked Solution: Administration of progestogens or progestogen–estrogen combinations or IUDs within 72 hours of coitus have been found to be very effective as emergency contraceptives to avoid possible pregnancy due to rape or casual unprotected intercourse.

BOSONANCE

Practice Question 14

Single-Correct MCQ

Why is periodic abstinence practiced specifically from **day 10 to 17** of the menstrual cycle?

(1) Because menstruation occurs during days 10 to 17.

(2) Because ovulation is expected during this period and chances of fertilisation are very high (fertile period).

(3) Because the corpus luteum degenerates on day 10.

(4) Because cervical mucus completely blocks sperm entry only between days 10 and 17.

Correct Answer: (2)

Worked Solution: In periodic abstinence, couples abstain from coitus from day 10 to 17 of the menstrual cycle when ovulation could be expected. Since the chances of fertilisation are very high during this window, it is called the fertile period.

7. Common NEET UG Pitfalls & Rapid Revision Checklist

High-Frequency Exam Pitfalls to Avoid

- **Pitfall 1 — Confusing IUD Classes:** Students frequently mix up *Lippes loop* (non-medicated), *CuT / Cu7 / Multiload 375* (copper-releasing), and *Progestasert / LNG-20* (hormone-releasing). Memorise each example under its exact class.
- **Pitfall 2 — ZIFT vs. IUT Blastomere Threshold:** Up to 8 blastomeres goes to the **fallopian tube (ZIFT)**; **more than 8 blastomeres** goes to the **uterus (IUT)**. Never reverse this boundary.
- **Pitfall 3 — GIFT vs. ZIFT:** In **GIFT**, an unfertilised *gamete* (donor ovum) is transferred into the fallopian tube, so fertilisation happens *in vivo*. In **ZIFT**, a fertilised *zygote/early embryo* formed *in vitro* is transferred into the fallopian tube.
- **Pitfall 4 — Sterilisation vs. Gametogenesis:** Vasectomy and tubectomy block **gamete transport**, NOT gamete formation or sex hormone secretion.
- **Pitfall 5 — Reusable vs. Disposable Barriers:** Male and female condoms are **disposable**; diaphragms, cervical caps, and vaults are **reusable**.
- **Pitfall 6 — Year of Family Planning vs. MTP Legalisation:** Family planning programmes were initiated in India in **1951**; MTP was legalised in India in **1971** (and amended in **2017**).

Pre-Exam Revision Checklist

- ✓ Can you state the WHO definition of reproductive health (physical, emotional, behavioural, and social well-being)?
- ✓ Do you know the genetic disorders tested by amniocentesis (Down syndrome, haemophilia, sickle-cell anemia) and why sex determination by amniocentesis is banned?
- ✓ Do you remember key dates and numbers: 1951 (Family Planning), 1971 (MTP legalised), 2017 (MTP Amendment Act), 18 yrs (female marriageable age), 21 yrs (male marriageable age), $< 2\%$ or $20/1000/\text{year}$ (2011 census growth rate)?
- ✓ Can you distinguish the three natural methods: Periodic abstinence (days 10–17), Withdrawal (*coitus interruptus*), and Lactational amenorrhea (max 6 months)?

BOSONANCE

- ✓ Do you know the regimen for standard oral pills (21 days daily starting within first 5 days of cycle + 7-day gap) versus *Saheli* (non-steroidal, once-a-week, developed at CDRI Lucknow)?
- ✓ Can you list the three incurable STIs (Hepatitis-B, Genital herpes, HIV) and the vulnerable age group (15–24 years)?
- ✓ Can you match every ART abbreviation (IVF–ET, ZIFT, IUT, GIFT, ICSI, AI, IUI) to its exact procedure and indication?

8. Concise Final Recap

Chapter Summary in 6 Key Points

1. **Reproductive Health & RCH:** Total well-being in physical, emotional, behavioural, and social aspects of reproduction. India launched Family Planning in 1951, now operating as Reproductive and Child Health Care (RCH) programmes.
2. **Population & Contraception:** Rapid decline in death rate, MMR, and IMR drove population growth. Contraceptive options include Natural methods, Barriers (condoms also prevent STIs), IUDs (non-medicated, Cu-releasing, hormone-releasing), Oral Pills (steroidal 21-day pills and non-steroidal weekly *Saheli*), Injectables/Implants, Emergency contraceptives (within 72 hours), and Terminal Surgical Sterilisation (Vasectomy and Tubectomy).
3. **MTP:** Legalised in India in 1971 (amended in 2017); accounts for $45–50\ \text{million}$ procedures annually worldwide ($1/5\text{th}$ of conceived pregnancies). Safe up to 12 weeks (1st trimester; 1 doctor's opinion); between 12 and 24 weeks requires 2 doctors' opinions on specified medical/foetal grounds.
4. **STIs:** Transmitted sexually (Hepatitis-B and HIV also via needles, blood transfusion, and mother-to-foetus). All are curable if detected early **except Hepatitis-B, Genital herpes, and HIV**. Untreated STIs lead to PID, ectopic pregnancy, still birth, infertility, and reproductive tract cancer.
5. **Infertility & ART:** Inability to conceive after 2 years of unprotected cohabitation. Treated via IVF–ET (ZIFT ≤ 8 blastomeres into fallopian tube; IUT > 8 blastomeres into uterus), GIFT (donor ovum into fallopian tube), ICSI (direct sperm injection into ovum), and AI/IUI (semen transfer into vagina/uterus for low sperm count or insemination inability).
6. **Legal Adoption:** Because ART requires expensive instrumentation and high precision and is accessible to few, legal adoption of orphaned and destitute children remains one of the best methods for couples seeking parenthood.

BOSONANCE